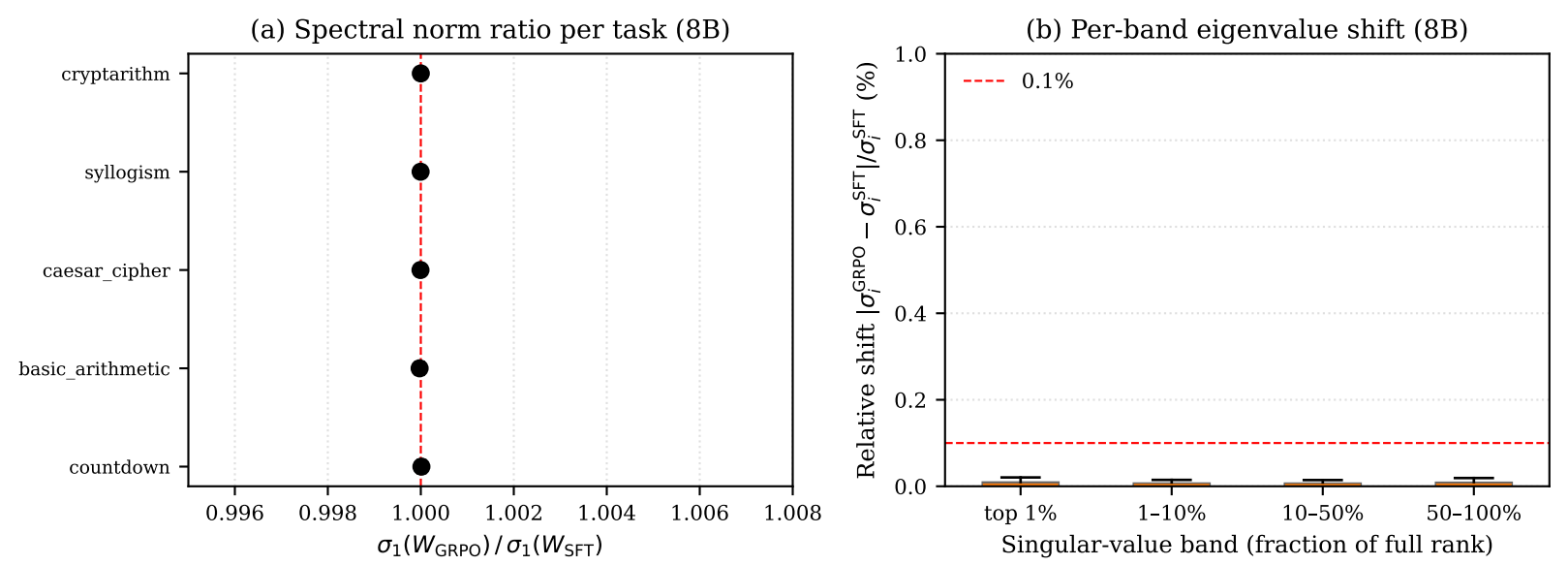
(a) Spectral norm ratio per task (8B)
cryptarithm
syllogism
caesar_cipher
basic_arithmetic
countdown
0.996
0.998
1.000
1.002
1.004
1.006
1.008
$\sigma_1(W_{\mathrm{GRPO}}) / \sigma_1(W_{\mathrm{SFT}})$
(b) Per-band eigenvalue shift (8B)
0.1%
Relative shift $|\sigma_i^{\mathrm{GRPO}} - \sigma_i^{\mathrm{SFT}}| / \sigma_i^{\mathrm{SFT}}$ (%)
0.0
0.2
0.4
0.6
0.8
1.0
top 1%
1–10%
10–50%
50–100%
Singular-value band (fraction of full rank)